

```
# Author: Cameron McGhee
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.5.1
```

```
## v ggplot2    4.0.0      v tibble     3.2.1
```

```
## v lubridate  1.9.4      v tidyr      1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(boot)
```

```
library(scales)
```

```
##
```

```
## Attaching package: 'scales'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##      discard
```

```
##
```

```
## The following object is masked from 'package:readr':
```

```
##
```

```
##      col_factor
```

```
box_path <- "NBA2024.csv"
```

```
def_path <- "team_def_rating_2024.csv"
```

```
box <- read_csv(box_path, show_col_types = FALSE)
```

```
def <- read_csv(def_path, show_col_types = FALSE)
```

```
league_def_avg <- def$opp_def_rating %>% mean(na.rm = TRUE)
```

```
box <- box %>%
```

```
  left_join(def, by = "opponent") %>%
```

```
  mutate(
```

```
    MissedFG = pmax(FGA - FGM, 0),
```

```
    OppEffAdj = league_def_avg / opp_def_rating
```

```
  )
```

```
compute_poss_used <- function(df) {
```

```
  df %>% mutate(
```

```
    Poss_used = FGA + 0.44*FTA + TOV + 0.2*AST + 0.5*OREB + 0.5*(STL + BLK),
```

```
    Poss_used = if_else(Poss_used <= 0, NA_real_, Poss_used)
```

```
  )
```

```
}
```

```
compute_IER <- function(df) {
```

```
  df %>%
```

```

  compute_poss_used() %>%
  mutate(
    IER_raw = (PTS + 1.5*AST + 1.2*OREB + 1.3*STL + 1.5*BLK - 1.2*TOV - 0.8*MissedFG - 0.5*PF) / Poss,
    IER = IER_raw * OppEffAdj
  )
}

ier_df <- compute_IER(box)

boot_IER_player <- function(player_row, R = 1000) {

  set.seed(42)
  comps <- c("PTS", "AST", "OREB", "STL", "BLK", "TOV", "MissedFG", "PF", "FGA", "FTA")
  vals <- as.numeric(player_row[comps])
  names(vals) <- comps
  OppEffAdj <- as.numeric(player_row["OppEffAdj"])
  if (any(is.na(vals)) || is.na(OppEffAdj)) return(c(NA, NA, NA))

  sim_IER <- numeric(R)
  for (r in seq_len(R)) {
    sim <- rpois(length(vals), lambda = pmax(vals, 0))
    names(sim) <- comps
    Poss_used <- sim["FGA"] + 0.44*sim["FTA"] + sim["TOV"] + 0.2*sim["AST"] + 0.5*sim["OREB"] + 0.5*(sim["STL"] + sim["BLK"])
    if (Poss_used <= 0) {
      sim_IER[r] <- NA_real_
    } else {
      IER_raw <- (sim["PTS"] + 1.5*sim["AST"] + 1.2*sim["OREB"] + 1.3*sim["STL"] + 1.5*sim["BLK"] - 1.2*sim["TOV"] - 0.8*sim["MissedFG"] - 0.5*sim["PF"]) / Poss_used
      sim_IER[r] <- IER_raw * OppEffAdj
    }
  }
  c(mean = mean(sim_IER, na.rm = TRUE),
    lo = quantile(sim_IER, 0.025, na.rm = TRUE),
    hi = quantile(sim_IER, 0.975, na.rm = TRUE))
}

boot_res <- ier_df %>%
  rowwise() %>%
  mutate(
    IER_boot_stats = list(boot_IER_player(cur_data()))
  ) %>%
  ungroup() %>%
  mutate(
    IER_boot_mean = map_dbl(IER_boot_stats, 1),
    IER_boot_lo = map_dbl(IER_boot_stats, 2),
    IER_boot_hi = map_dbl(IER_boot_stats, 3)
  ) %>%
  select(-IER_boot_stats)

```

```

## Warning: There was 1 warning in `mutate()`.
## i In argument: `IER_boot_stats = list(boot_IER_player(cur_data()))`.
## i In row 1.
## Caused by warning:
## ! `cur_data()` was deprecated in dplyr 1.1.0.

```

```
## i Please use `pick()` instead.
cor_results <- list()
if (all(c("PER", "WS48") %in% names(boot_res))) {
  cor_results$IER_PER <- cor(boot_res$IER, boot_res$PER, use = "complete.obs")
  cor_results$IER_WS48 <- cor(boot_res$IER, boot_res$WS48, use = "complete.obs")
} else if ("PER" %in% names(boot_res)) {
  cor_results$IER_PER <- cor(boot_res$IER, boot_res$PER, use = "complete.obs")
}
print(cor_results)

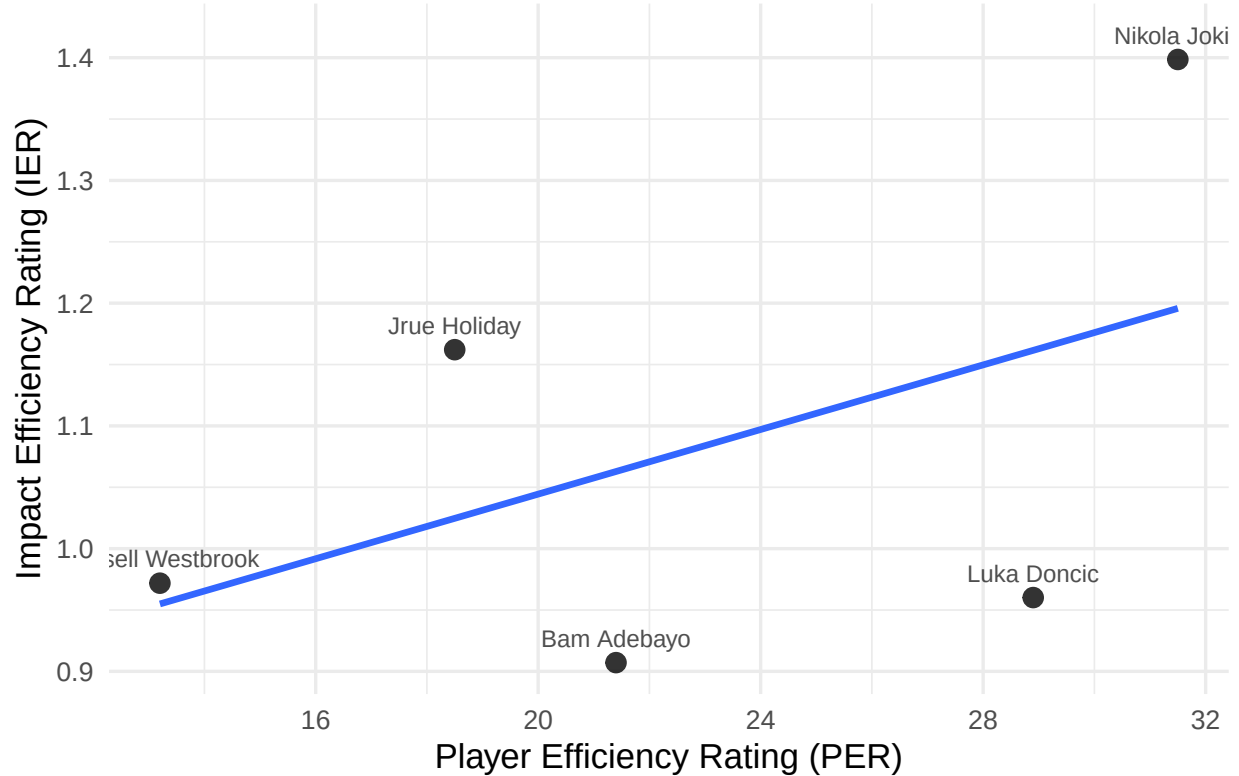
## $IER_PER
## [1] 0.4877384
##
## $IER_WS48
## [1] 0.5904061

# Figure 1: PER vs IER
if ("PER" %in% names(boot_res)) {
  p1 <- ggplot(boot_res, aes(x = PER, y = IER, label = player)) +
    geom_point(size = 3, alpha = 0.8) +
    geom_smooth(method = "lm", se = FALSE) +
    geom_text(nudge_y = 0.02, size = 3, alpha = 0.7) +
    labs(title = "Comparing IER and PER",
         x = "Player Efficiency Rating (PER)",
         y = "Impact Efficiency Rating (IER)") +
    theme_minimal(base_size = 13)
}
p1

## `geom_smooth()` using formula = 'y ~ x'

## Warning: The following aesthetics were dropped during statistical transformation: label.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
## variable into a factor?
```

Comparing IER and PER



```
# Figure 2: Top players by IER with 95% CI
top_n <- 10
p2 <- boot_res %>%
  arrange(desc(IER)) %>%
  slice_head(n = top_n) %>%
  ggplot(aes(x = reorder(player, IER), y = IER)) +
  geom_col(alpha = 0.9) +
  geom_errorbar(aes(ymin = IER_boot_lo, ymax = IER_boot_hi), width = 0.2) +
  coord_flip() +
  labs(title = "Top Players by Impact Efficiency Rating (IER)",
       x = "Player",
       y = "IER (with 95% bootstrap CI)") +
  theme_minimal(base_size = 13)

p2
```

